



HealthConnect Appointment No-Show Risk

Predicting which appointments may need extra attendance support

Presenter

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Track

Data Science, AnalystLab Africa Experience Lab

Project

Improving Patient Appointment Attendance and Healthcare Support
Using Data and AI

Problem

Missed appointments waste clinic capacity and reduce opportunities to support patients before their visit

The Problem and My Role

HealthConnect Clinic Wants To

- Reduce missed appointments
- Improve attendance
- Make better use of appointment slots
- Provide more effective administrative support

📄 The model supports staff decisions — it does not automate them.

My Data Science Responsibility

Develop, test and prepare a predictive model that identifies appointments that may benefit from additional attendance support.

What I Produced

A final **38-feature Logistic Regression model** that outputs the probability of a No-Show for each appointment.

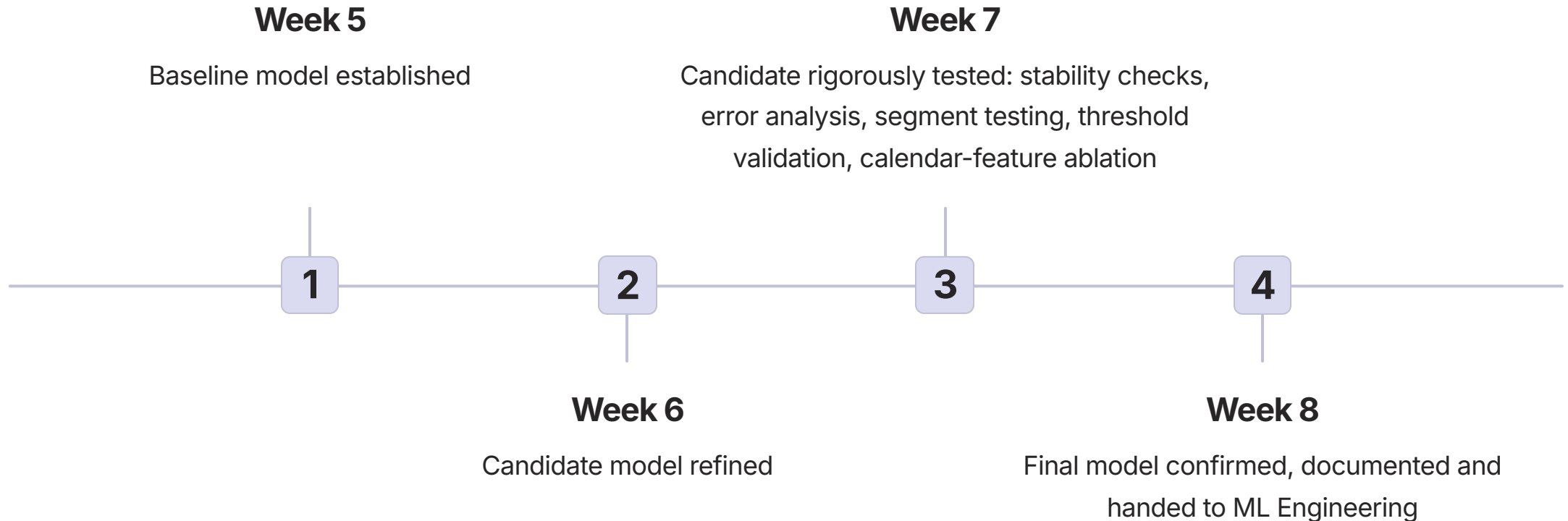
Methods and Tools

Python, pandas, scikit-learn, cross-validation, holdout evaluation, error analysis, ablation testing, joblib model artifact



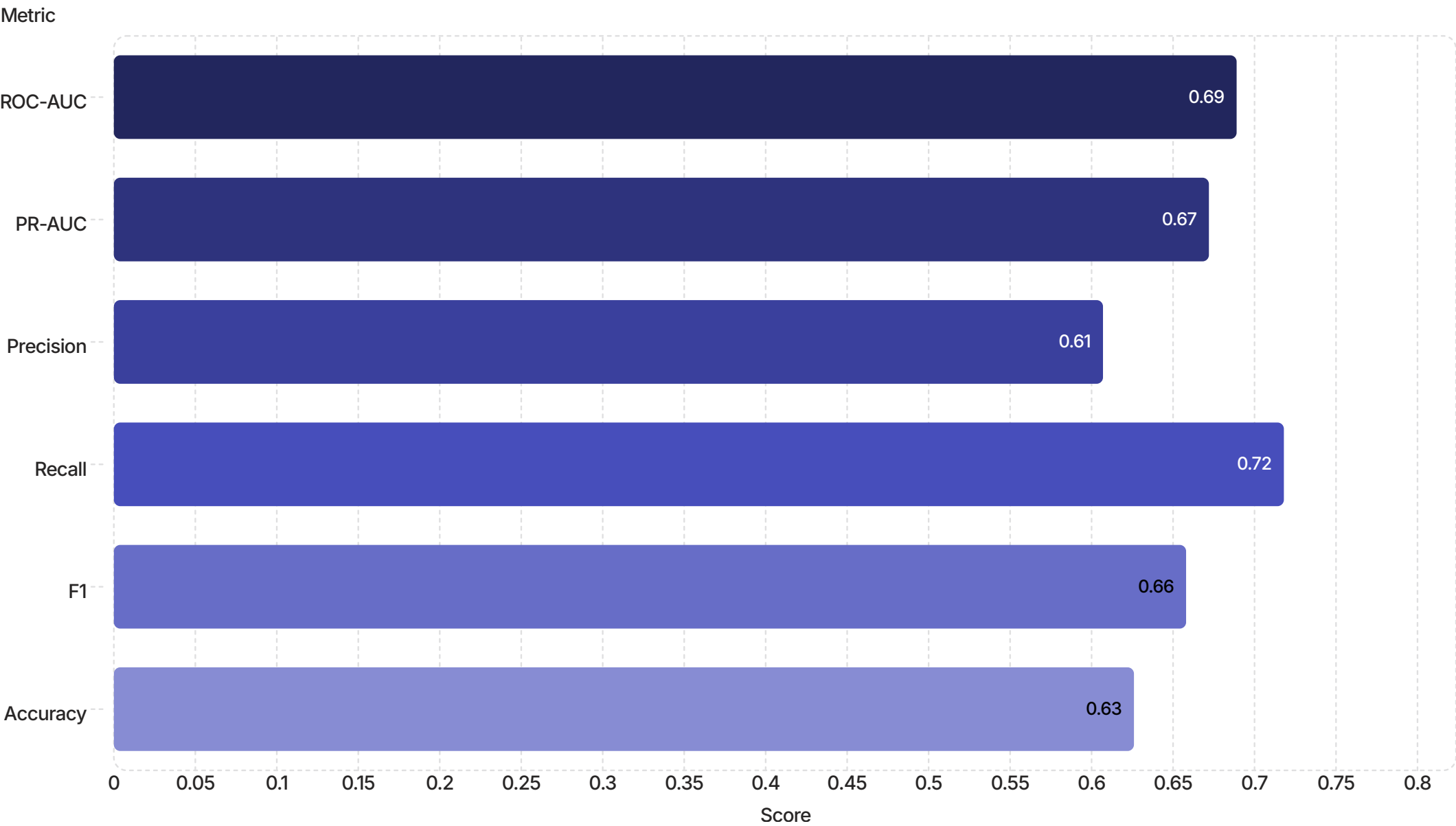
The Development Journey

Week 7 challenged the candidate rather than assuming it was an improvement.



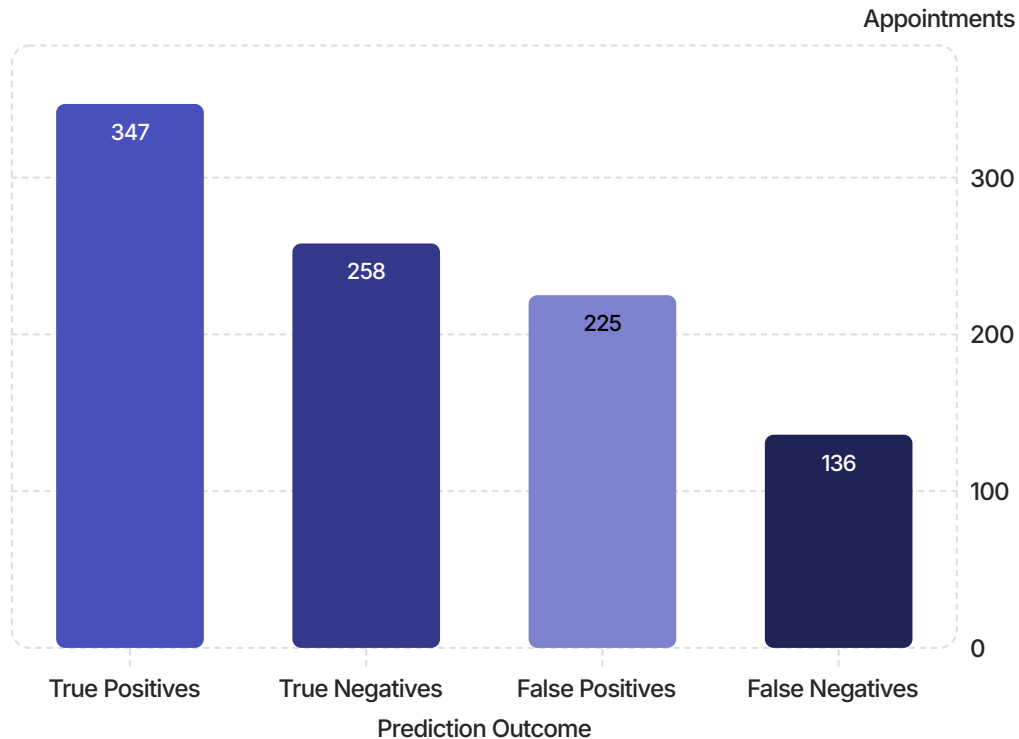
The Final Model: Performance on Holdout Data

Model: Logistic Regression with 38 features



❏ Performance is moderate, not perfect. The model is a useful risk-prioritisation signal, not a highly accurate predictor.

The Operational Trade-off at the 0.45 Threshold



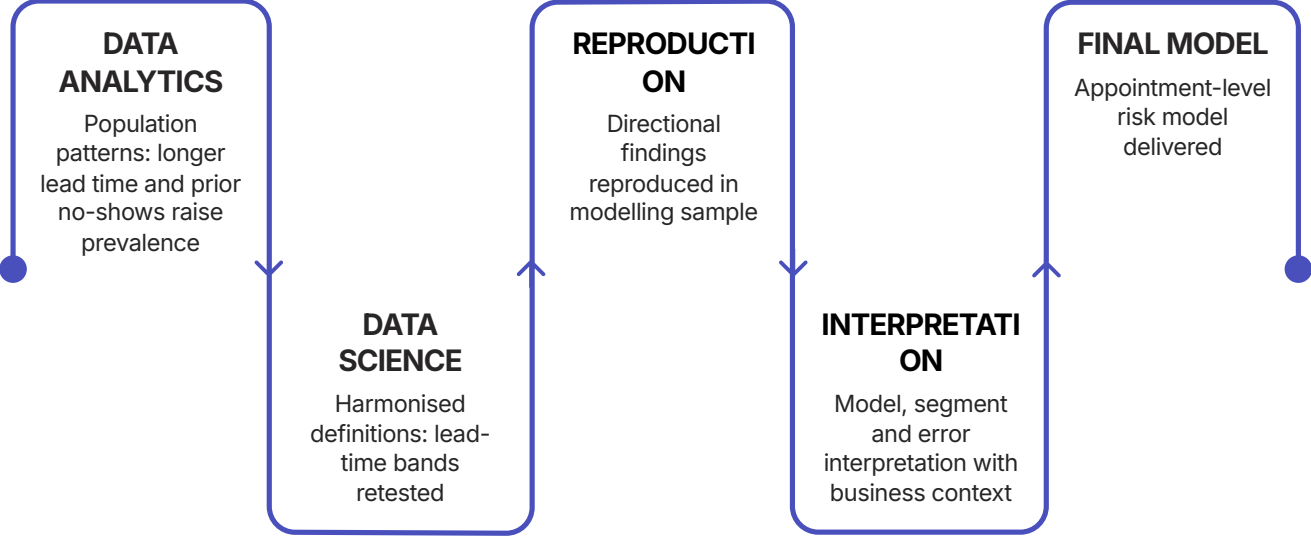
What Each Outcome Means

- **True Positives (347):** No-shows correctly flagged for outreach
- **True Negatives (258):** Attenders correctly left unflagged
- **False Positives (225):** Attenders flagged — extra outreach sent unnecessarily
- **False Negatives (136):** No-shows missed — no outreach triggered

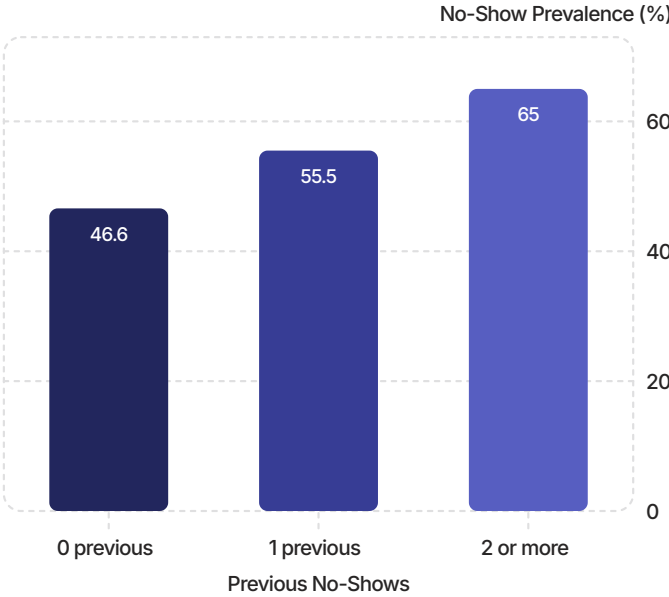
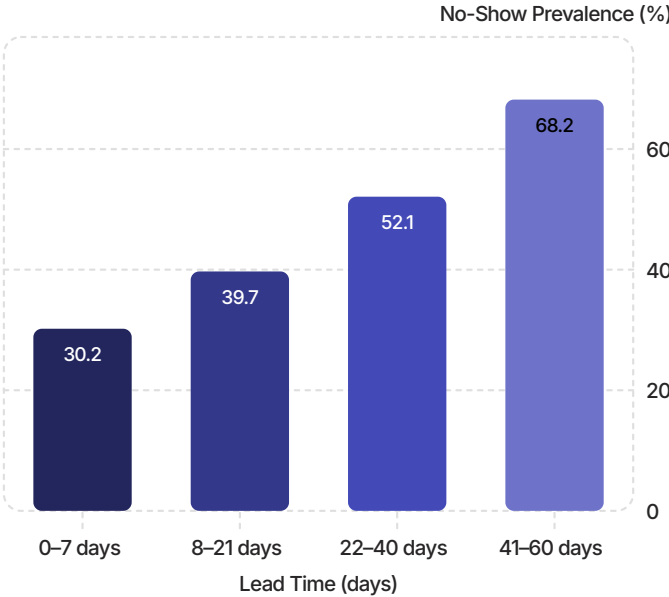
Increasing recall helps identify more potential no-shows, but it also increases outreach to patients who would have attended.

Threshold selection is a business decision as well as a modelling decision. The 0.45 threshold remains provisional.

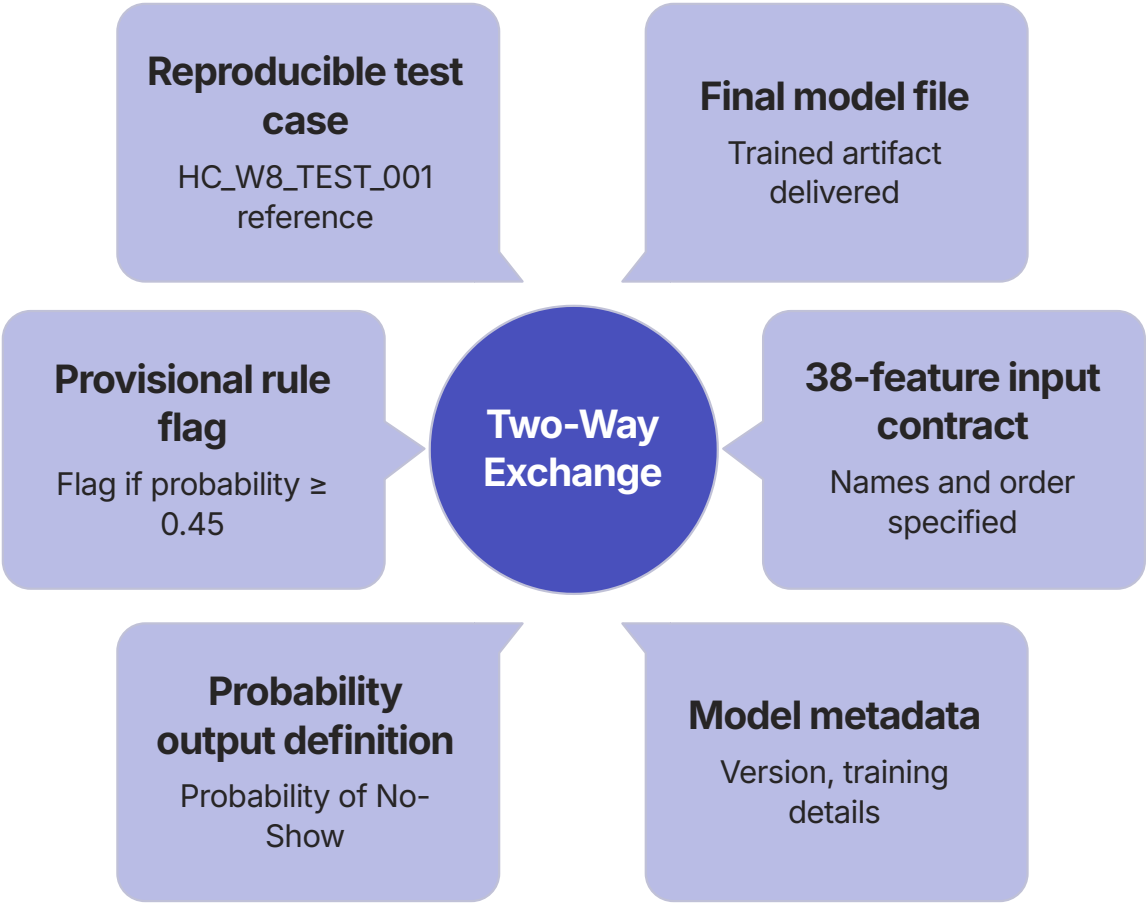
Cross-Track Collaboration: Data Analytics to Data Science



Lead-time definitions were aligned across tracks before retesting. These are associations, not causes.



Cross-Track Collaboration: Data Science and ML Engineering

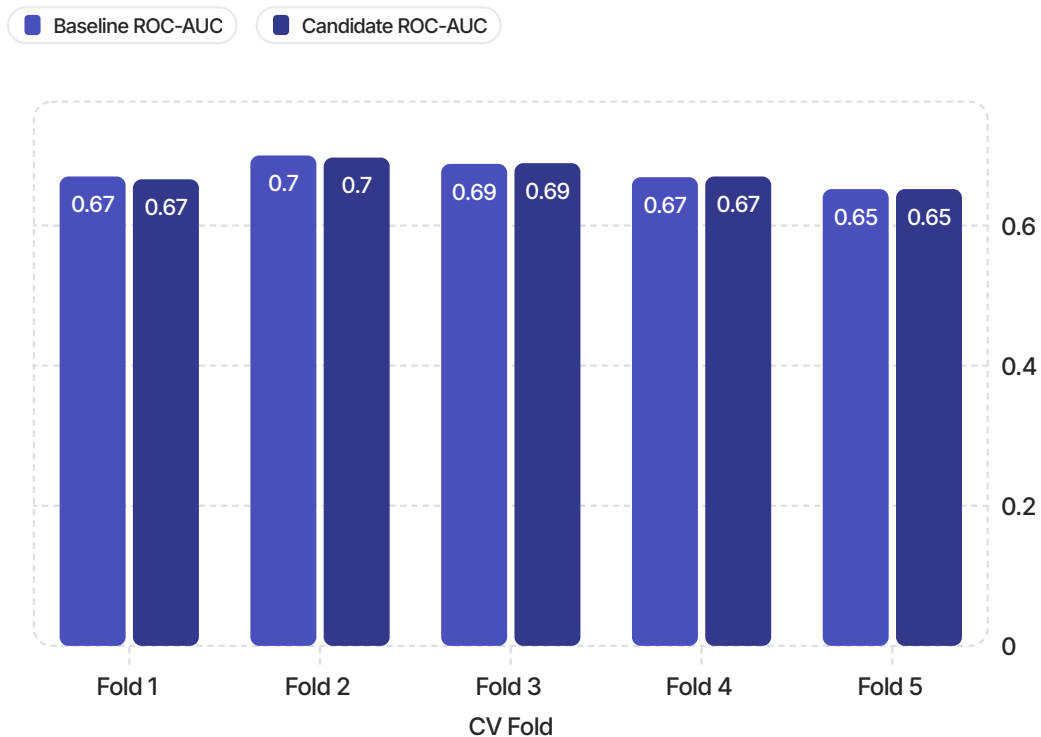


Reference Test:
HC_W8_TEST_001

Expected Probability
0.4951596731
Risk Flag
1 (Higher Risk)
Tolerance
0.000001

⚠ Honest status: handoff completed and pre-handoff checks passed. The pipeline result has not yet been returned, so end-to-end reproduction is **PENDING** and is not reported as validated.

Testing and Refinement



Cross-Validation Result

The candidate won **3 of 5 folds** on ROC-AUC and **2 of 5** on PR-AUC. Mixed improvement — no exaggerated performance claim is made.

Calendar-Feature Ablation

Simplification tested by removing day-of-year and month. Recall at 0.45 fell by **0.0021**, narrowly exceeding the predefined **0.002 tolerance**, so the simplification was rejected and the full 38-feature model retained.

📌 Lesson: decisions followed predefined rules, not hopes.

The Overall HealthConnect Solution: Value and Limits



✓ Value

- Risk prioritisation of appointments
- Targeted supportive outreach
- Interpretable probability output
- Better use of limited staff time

⚠ Limitations

- Moderate discrimination (ROC-AUC 0.689)
- False-positive and false-negative trade-off
- Provisional threshold
- Not production-ready
- Associations are not causes

🎯 Appropriate Use

- Prioritise reminders and outreach
- **NOT** for denying access
- **NOT** for cancelling appointments
- **NOT** for penalising patients
- **NOT** for clinical decisions

How my work supports HealthConnect: it turns population patterns into appointment-level risk so staff can focus support where it may matter most.

Key Takeaways and Next Steps

Key Takeaway

A transparent, honestly evaluated model can guide supportive outreach when its limits are clearly communicated.

What I Learned

- Rigorous testing and honest reporting matter more than headline metrics
- Cross-track collaboration changes outcomes — aligning definitions before retesting made a real difference

Next Steps for a Future Version

01

Complete ML Engineering pipeline validation

02

Obtain operational approval of the threshold

03

Set up monitoring for drift

04

Resolve patient-identifier traceability

05

Use patient-grouped and fully nested validation

Thank you